Stathis **Galanakis**

RESEARCH ENGINEER | MULTIMODAL GENERATIVE AI | MACHINE LEARNING

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Summary

Research Engineer (PhD, Imperial College London) specialising in generative AI, diffusion models, and multimodal machine learning systems. My work focuses on scalable generative modeling under low-data and real-world constraints, with applications spanning synthetic data generation, representation learning, and identity-consistent generation. Published first-author research at ICCV and WACV, with experience building large-scale datasets, training pipelines, and generative ML systems across academia and industry.

Experience

Research Associate in Generative Computer Vision Models — London, UK
Department of Computing, Imperial College London — *March 2025 - Present*

- Developed rectified flow-based image synthesis for low-data augmentation, resulting in an 8% improvement in pathology classification.
- Scaled training of high-resolution generative models by deploying multi-GPU workflows (PyTorch DDP).
- Engineered LLM workflows for synthetic medical caption generation (500k image-text pairs) and explored automated pathology report classification.
- Supervised Master's students on research projects in generative AI and world modeling.

Research Engineer (Internship) — London, UK
Huawei UK — *January 2022 - March 2025*

- Built and evaluated diffusion-based generative models for robust identity-preserving image synthesis under ambiguous visual conditions, achieving sota identity preservation performance.
- Curated synthetic identity datasets (10k+ identities) to improve model robustness and generalisation.
- Designed neural rendering pipelines for controllable multi-view avatar generation and consistent viewpoint synthesis, across varying expressions, poses and lighting conditions.

Research Assistant — London, UK
Business School, Imperial College London — *February 2021 - January 2022*

- Applied machine learning to multimodal satellite imagery and weather data for crop yield prediction under spatial data sparsity.
- Developed synthetic data augmentation strategies to improve generalisation in underrepresented geographic regions.

Computer Vision Scientist — London, UK
ArielAI (acquired by Snap Inc.) — *January 2020 - September 2020*

- Built large-scale data collection and filtering pipelines for training vision models, incorporating human-in-the-loop annotation to improve dataset quality.
- Contributed to scalable data infrastructure supporting model development in a startup environment, later acquired by Snap Inc.

R&D, ML Engineer — Athens, Greece
Pobuca Ltd — *May 2018 - January 2019*

- Designed and deployed an end-to-end product detection system for retail shelf images (25+ classes), integrating semi-automated annotation and training workflows.

Technical Strengths

Programming:	Python, CUDA, C++, Linux, Git, Docker
Generative AI:	Diffusion Models, Video Diffusion, Rectified Flows, Transformers
3D & Vision:	3D Face Reconstruction, 3D Scene Reconstruction, NeRF, 3D Gaussian Splatting (3DGS)
Frameworks:	PyTorch, PyTorch Lightning, PyTorch3D, Accelerate
Systems:	Distributed Training, Multi-GPU Scaling, SLURM, Large-scale Data Pipelines

Selected Publications

DermaFlux: Synthetic Skin Lesion Generation with Rectified Flows for Enhanced Image Classification
Galanakis S, Koliousis A, and Zafeiriou S. — MICCAI 2026

DermaFlux is a rectified flow–based text-to-image model for synthetic skin lesion image synthesis.

STARCaster: Spatio-Temporal AutoRegressive Video Diffusion for Identity- and View-Aware Talking Portraits
Paraperas F, **Galanakis S**, Potamias R, Kainz B, and Zafeiriou S. — ICML 2026

SpinMeRound: Consistent Multi-View Identity Generation Using Diffusion Models
Galanakis S, Lattas A, Moschoglou S, Kainz B, and Zafeiriou S. — ICCV 2025

SpinMeRound introduces a diffusion framework for identity-consistent multi-view visual generation from sparse inputs.

ImHead: A Large-scale Implicit Morphable Model for Localized Head Modeling
Potamias R, **Galanakis S**, Deng J, Papaioannou A, and Zafeiriou S. — ICCV 2025

FitDiff: Robust monocular 3D facial shape and reflectance estimation using Diffusion Models
Galanakis S, Lattas A, Moschoglou S, and Zafeiriou S. — WACV 2025

FitDiff introduces a multimodal diffusion framework for robust identity-preserving visual generation from unconstrained images.

3DMM-RF: Convolutional Radiance Fields for 3D Face Modeling
Galanakis S, Gecer B, Lattas A, and Zafeiriou S. — WACV 2023

3DMM-RF explores neural radiance field representations for controllable and identity-consistent visual generation, under diverse expressions, poses and lighting conditions.

Education

PhD in Computer Science — Imperial College London, UK
Supervisor: *Prof. Stefanos Zafeiriou* — April 2025

Thesis: Advancements in Face Reconstruction from a Single Image.
Research focused on diffusion-based generative modeling and robust visual generation under unconstrained real-world conditions. Work resulted in peer-reviewed publications (ICCV, WACV), including first-author contributions.

Diploma/M.Eng. in Electrical and Computer Engineering — NTUA, Greece
Supervisor: *Prof. Petros Maragos* — November 2019

Thesis: Human Action Recognition and Localisation in Videos.

Interests

· Reviewer for international conferences such as CVPR (2024, 2025), ECCV (2026) and MICCAI (2026).

· Passionate about basketball, cycling, and running.